

Getting Back Together: HCI and Human Factors Joining Forces to Meet the AI Interaction Challenge

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ABSTRACT

The Human Factors Society and ACM SIGCHI jointly organized the first CHI conference in 1983, but during the remainder of the 1980s, Human-Computer Interaction (HCI) and Human Factors Engineering (HFE) increasingly diverged. The focus of HCI shifted from exploring systems for routinized activities of trained personnel, to a more general use of technology. HCI became predominantly design-oriented, concentrating on usability and user experience, moving further from HFE principles. However, the rapid growth of Artificial Intelligence (AI) applications posed unique and urgent challenges that call for a reestablishment of the connection between the two disciplines. We argue that by working as a team, HCI and HFE could more effectively address AI-posed challenges. We invite HCI and HFE researchers to take part in a full-day interactive hybrid workshop. With this workshop, we aim to initiate a collaboration between HCI and HFE and set a clear plan forward for a more united future.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); • **Social and professional topics** → Professional topics; History of computing; • **Computing methodologies** → Artificial intelligence.

KEYWORDS

Human-Computer Interaction (HCI), Human Factors Engineering (HFE), Artificial Intelligence (AI), Cognitive Science, human augmentation, human-AI interaction, human-centered computing

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1 WORKSHOP MOTIVATION

In the early days of computing, Human-Computer Interaction (HCI) interests were closely aligned with those of Human Factors Engineering (HFE). HCI research was predominantly focused on routinized activities by people, usually trained personnel, who would be users of costly systems not used for discretionary purposes [9]. HCI researchers explored ergonomics, management information systems [9], as well as HFE topics on personnel selection, training standards and guidelines [6]. However, with the introduction of the personal computer in the early 1980s, the non-discretionary role of the operator or a trained worker was replaced primarily with the user role [9]. The waves of technological advances (e.g., the world of the wide web, ubiquitous computing, video conferencing, and advanced robotics) have redefined the meaning and capabilities of user interfaces, creating an increased emphasis on iterative design and less emphasis on applied psychology and theory-based design. In response to the introduction of the personal computer and the waves of technological change since then, the field of HCI has become predominantly design-oriented, concentrating on usability and the more general concept of user experience. Thus the discipline of HCI currently receives relatively little guidance from HFE principles, ergonomics, and applied psychology approaches [6]. While its primary emphasis has been on technological and methodological innovation, invention, and design [6], earlier concepts of human-machine symbiosis [11] and human augmentation [8], are becoming more relevant with the growth of AI applications and the need to design effective human-AI interactions. Thus, to cope with the human-AI interaction challenge, the four decades of separation between HFE and HCI must now be reconsidered. New HCI challenges posed by the advances in AI create an opportunity to re-explore synergies between AI and HFE. Many of the critical issues that the rise of AI has brought are traditional HFE problems. HFE has spent decades focusing on supervisory control, trust in automation, situation awareness, mental workload assessment, and interruptibility, for example. Re-establishing the connection between the two disciplines and encouraging the development of teams that combine HFE and HCI specialists could be particularly fruitful in addressing questions related to these concepts. For example, HCI working in tandem with HFE could more effectively address challenges related to explainable AI, which is an area that allows people to understand the underlying reasoning of machine learning predictions so they can build appropriate trust in the results [5, 12, 16]. HFE could also inform HCI on issues related to active learning by bringing some insights on the well-explored

subject of interruptability (e.g., [1, 2]). Moreover, combining the strengths of both disciplines could also help resolve some interface design issues. An increased ability to interface with machine processes could augment people's ability to synthesize data and make effective and efficient decisions [6].

Bridging the disciplinary divide is also increasingly important in the rise of HCI's human-centered focus [3, 15]. Re-establishing the collaborative connection between the two disciplines could enrich the development of Human-centered AI systems. HCI and HFE working together would create synergies between their respective strengths. For example, the HCI approach has focused on understanding users through a predominantly empirical process. That approach uses methods such as contextual inquiry or surveys to conduct research before developing, applying, and testing different styles and instantiating interfaces to machine learning. At the same time, the HFE approach has typically focused on theory-based analysis of AI issues, particularly cognitive science and computational theory [6]. Thus, HFE could provide theoretical support from cognitive science and views of human performance, while HCI provides necessary design tools and methods. This could also, in part, address the criticism that by separating from HFE and applied psychology approaches, HCI lost some clarity about its grounding in science [4, 7, 10, 13, 14]. As in any reconciliation process, the goal of this workshop will not be to "rake over the old coals" of the former HFE-HCI relationship and the relative merits of their approaches and scientific backing but instead to focus on the challenge of human-AI interaction, which makes the prospect of reconciliation both necessary and exciting.

The seminal paper by Chignell et al. [6] offers insights to provide a new outlook on the respective roles of HCI and HFE in the past and future as they face the challenge of designing safe, efficient, and comfortable human-AI interaction. Chignell and Sellen met at CHI 2024 in Hamburg and formulated the goal bringing HFE and CHI back together again to address the challenge of human-AI interaction. Our goal in planning this workshop was not to steer HCI in a different direction. Instead, we want to promote a rapprochement between the fields of HCI and HFE that they jointly address the challenges and opportunities created through applications where humans interact with AI. We suggest that HCI could benefit by re-engaging with HFE branches exploring human-machine system integration that are relevant to the application of HCI in human-AI interaction. This re-engagement could be fruitful for those sub-topics of common interest in which both fields can offer complementary contributions.

This workshop aims to encourage a discussion about broadening HCI curricula to include content relating to human-AI interaction. Without disputing that HCI is addressing the issues well-studied by HFE, we argue that the two fields will make more progress working together. We seek to invite researchers and practitioners to build guidance on how this collaboration could be achieved after 40 years of growing apart. With the new opportunities presented by AI, it is time to consider bringing HCI and HFE researchers and practitioners back together again ("for the sake of the AI kids").

This workshop is also an opportunity for HCI and HFE research community members to jointly raise, brainstorm, and address the challenges and opportunities created through applications where humans interact with AI. For example, how can we build inclusivity

more directly into HCI by elevating the role of human augmentation as a fundamental goal in designing interactions? How could the two disciplines address the questions of augmentation of human skills with AI? Human augmentation will require considering non-discretionary (e.g., automated vehicles, clinical decision support) and discretionary (e.g., gaming, collaboration over social media) situations. Given that people are already diverse, different levels of augmentation will add diversity on top of diversity. In addition to diverse forms of augmentation, how can we handle so many types of diversity - diversity of abilities, age, personality, and motivation, just to name a few? Can a more inclusive view of HCI as human augmentation make for a more equitable and effective discipline? This workshop will invite researchers to consider these questions and foresee challenges that will require collaborative approaches in the future.

Finally, this workshop is intended to build a new research community and establish a collaborative relationship between the members of HCI and HFE. Despite working on increasingly similar challenges of human-AI interaction, human augmentation, and inclusive design, members of the two fields are notoriously unconnected. Since the ACM SIGCHI conference 1983, which was run jointly, for the first and only time, by ACM and the Human Factors Society, HFE and HCI researchers often publish in different venues and attend different conferences. Even when working in the same institution, HCI and HFE members are often housed in different academic and industry departments. This workshop is intended to be a first step towards a more connected research community that could trust and rely on each other to collaborate on the design of augmented humans, vehicle automation, and other topics of common interest (where topics such as trust and reliance are of particular importance).

2 ORGANIZERS

Alyssa Iglar is a PhD Student in the Department of Mechanical and Industrial Engineering at the University of Toronto. Alyssa specializes in Human Factors and is pursuing her doctoral research in the area of implementation science, specifically in relation to technology-centered implementation projects and sustained use of these technologies in long-term care settings. She currently serves as president of the Human Factors Interest Group (HFIG), the University of Toronto's student chapter affiliated with the Human Factors and Ergonomics Society.

Auste Simkute is a PhD Candidate in Design at the University of Edinburgh. Her research explores the potential applications of HFE in explainable AI-driven decision-making in expert domains. Auste aims to combine HCI, HFE, and Cognitive Psychology knowledge to develop design guidelines for tailored explainability, enabling experts to effectively use their decision-making strategies and benefit from using AI-driven technologies.

Abigail Sellen is Distinguished Scientist and Lab Director at Microsoft Research Cambridge in the UK. Abigail is a Fellow of the Royal Society, the Royal Academy of Engineering, the ACM, and member of the CHI Academy. Her early career was in Human Factors engineering (and taught by Chignell!) but she veered off into HCI after a PhD with Don Norman, and an internship at Apple Computer in the late '80s. She identifies mainly as an HCI researcher but feels nostalgic about her HFE past.

Mark Chignell is a Professor in the Department of Mechanical and Industrial Engineering, University of Toronto. He is a member of both the human factors and information engineering groups in his department and he identifies as both an HFE and HCI researcher. He is working on technologies for aging including physical and cognitive exercises for people with dementia. He also does research on interactive machine learning in cybersecurity and healthcare and he is interested in the development of user interfaces for communicating efficiently and effectively with artificially intelligent agents.

3 PLANS TO PUBLISH WORKSHOP PROCEEDINGS

Prospective participants will be asked to submit initial position papers that are 2–4 pages in length. These submissions are expected to address or comment on ways in which HCI and HFE methods and perspectives can work in tandem to address AI challenges, questions, and opportunities in general. Alternatively, authors may decide to focus on AI challenges, questions, and opportunities from one industry (e.g., aviation) and/or discuss either HCI or HFE methods and perspectives exclusively.

Authors of accepted submissions will be offered the opportunity to prepare manuscripts that are 15–20 pages in length which would subsequently be considered for inclusion as chapters in an edited book describing the workshop. If this workshop proposal is accepted, the workshop organizers will submit proposals to the Cambridge University Press, MIT Press, and Oxford University Press using the procedures outlined in [17] and [18], respectively.

Manuscripts submitted during both stages will be selected based on quality, coherence, and relevance to the workshop's overarching topic. Initial position papers will either be accepted or rejected. The subsequent 15–20 page position papers will be accepted as is, accepted with revisions, or rejected before inclusion in the edited book in the case of the former two decisions.

4 PRE-WORKSHOP PLANS

The call for participation will be disseminated via traditional SIGCHI communication channels (e.g., mailing lists) and within the network of the organizers' institutions. Our website URL (<https://hcihfetogether.wordpress.com>) is included in this call for participation and the site will be launched in early December if our proposal is accepted.

Targeted distribution to relevant HCI and HFE industry and academic groups will be via social media outreach and relevant mailing lists such as the Human Factors Interest Group's (HFIG) [19] and the University of Edinburgh Institute of Design Informatics [20]. We will also contact key researchers in the fields of HCI and HFE. A prime selection criterion will be to maximize the diversity and originality of the proposed ideas and balanced involvement from members representing HCI and HFE. The organizers also strive to compile a heterogeneous workshop participants list in terms of the career stage of participants.

4.1 Important dates

- Call for papers/participation: December 15th, 2023
- Participant submission deadline: February 19th, 2024

- Initial position paper decision notifications to participants: Ten days before the conference's Early Registration Deadline

5 HYBRID PLANS AND ASYNCHRONOUS ENGAGEMENT

A hybrid workshop offering in-person, remote, and asynchronous engagement opportunities will be hosted to allow for participation regardless of time zones, internet access, plans to attend CHI 2024 in person or remotely, and additional considerations. One organizer will attend the workshop online and moderate the Microsoft Teams channel that will be used to encourage participants attending online or engaging asynchronously to connect with each other as well as in-person participants, who will be welcome to use the channel as well. The workshop will be broadcast through the Microsoft Teams meeting associated with the channel. Additional methods and tools, as described in the subsequent "Workshop Activities" section, will be used to optimize engagement among remote and asynchronous participants, who we acknowledge can feel disengaged during hybrid events and require consideration during the planning of such initiatives.

On the day of the workshop, we will require a monitor or projector on which to display the Microsoft Teams meeting room and media for some of the activities (e.g., videos and presentation slides). One of the organizers will use their laptop to display this content and will require a compatible cable to connect their laptop to the monitor or projector. We might also need a microphone for participants and organizers attending in person to use during the workshop.

We are planning to have up to 30 in-person attendees and an additional 30 participate either synchronously or asynchronously online. All audio, presentation slides, and videos will be accompanied by captions and transcripts to enhance the accessibility of our workshop. Additionally, all organizer-facilitated tools that asynchronous participants might engage with will be compatible with screen readers, keyboards, and similar assistive devices. All physical materials used during the workshop will also be easy to read, handle, and use.

6 WORKSHOP ACTIVITIES

The workshop will include several interactive and engaging activities intended to facilitate thoughtful discussions centered around bringing HFE and HCI together in the age of AI will comprise this workshop. A primary intention for this workshop is to create a lively environment that encourages ideas and opinions from researchers in the HFE and HCI disciplines alike, as perspectives from both fields will be vital in paving the way for collaborative research and innovation to address challenges and questions posed by AI. The activities we are considering to include in the workshop are described below.

Keynote Addresses. Four distinguished speakers, each having expertise in HFE or HCI, will deliver 15-minute keynote addresses preceding and following the workshop's lunchbreak (one HFE and one HCI keynote speaker each before and after the lunchbreak). These keynote addresses are intended to provide additional context and perspective on areas of interest related to HFE, HCI, and/or AI,

which could inform or inspire participants' contributions during subsequent activities.

In addition to delivering their addresses in person, we will request that our keynote speakers pre-record their talks for the benefit of asynchronous participants. All participants will be welcome to ask questions about these addresses in the Microsoft Teams channel.

Lightening Talks. Authors of each accepted initial position paper will be asked to prepare a two-minute presentation (i.e., a lightening talk) summarizing their papers. These lightening talks, which are to be video recorded and shared with the workshop's participants and organizers for viewing before the workshop, are intended to align all parties on the opinions, perspectives, questions, and areas of expertise presented by and within each accepted workshop submission. Questions about each lightening talk may be posted online through the Microsoft Teams channel.

Responses to a Provocative Piece of Media. The workshop organizers will display a video or alternative form of media intended to elicit strong and/or opposing reactions from workshop participants in relation to the broader AI, HCI, and HFE-centered area of discussion and debate. This media piece will be posted through the Microsoft Teams channel for asynchronous viewing as well. Participants engaging in the workshop in person or synchronously online will be encouraged to share their immediate reactions to the media piece and propose accompanying key research issues, which will in turn fuel subsequent discussions and activities.

"Visions of the Future" Discussion Panel. Immediately after the provocative media activity, a select group of workshop participants (chosen based on their viewpoints expressed in the initial position papers) will form a panel and discuss participants' responses during the provocative media activity as well as research areas and directions relevant to the AI-centered issue or challenge presented through the media piece. Online asynchronous participants may choose to respond to the panel with a recorded video. The discussion panel, which will be moderated by the organizers, will provide a forum in which participants can understand and seek to resolve conflicting opinions amongst each other and begin collaborative idea generation in relation to future steps forward together in the AI domain.

Spectrogram Activity. After lunch, all workshop participants will be presented with a short series of provocative AI-centered statements (e.g., "no one will need a driver's license in the future", "software engineering will be obsolete in 10 years", "AI is an existential threat"). In-person participants will be asked to move to one side of the room if they agree with the statement and another side if they disagree. Online asynchronous and synchronous participants will be presented with a poll through which they may similarly express their views. In-person and online synchronous participants will subsequently be asked to elaborate on their decisions. Akin to the media-based activity, this exercise will provide a forum for attendees to learn about other participants' perspectives in relation to the growing and highly debated area of AI.

Clustering Activity. Attendees will engage in a clustering exercise in which in-person and online synchronous workshop participants alike will form smaller groups arranged by the organizers to brainstorm responses to the following question: What are the next big research challenges in relation to AI for HFE and HCI

experts to tackle together? Miro will be used to facilitate brainstorming among the in-person and online synchronous participants during the exercise. The activity is intended to encourage collaborative thinking among the HFE and HCI experts in attendance as well as inspire international collaborations (e.g., a new journal or alternative publication venue) following the workshop.

Closing Remarks. The workshop will conclude with the organizers' closing remarks, during which we will outline concrete actions that may be taken to bring the HCI and HFE communities together in addressing the challenges and questions posed by ai; describe how the publication of the final position papers and workshop outcomes will proceed; and inform participants of the mailing list exclusive to them that we will use for future communications.

7 POST-WORKSHOP PLANS

Contingent on acceptance of our proposal by the Cambridge University Press, MIT Press, or Oxford University Press, we will prepare an edited book detailing the outcomes of the workshop for future reference by those interested in the reunion of HCI and HFE for the sake of addressing AI's pressing challenges. Among the chapters of this book will be the final position papers submitted by the workshop participants and their co-authors. The final position papers will be received within three months of the workshop's conclusion, and all authors will be notified of the book's publisher (Cambridge University Press, MIE Press, or Oxford University Press) or whether the proposal was not accepted by any publisher. Asking authors of accepted submissions to expand upon their original position papers will allow for inspiration and insights from the workshop to be incorporated within the final publication, thereby producing an edited book that serves as a comprehensive and valuable discussion of the main topics considered during the workshop.

8 CALL FOR PARTICIPATION

Artificial intelligence (AI) and its accompanying challenges and questions calls for a reunion between the fields of human-computer interaction (HCI) and human factors engineering (HFE) – two previously intertwined disciplines marked by significant divergence over the last 30 years in terms of areas of interest and focus. The overarching goal of our workshop is to bring together experts from both fields and engage in spirited and collaboration-oriented discussions about the ever-evolving AI landscape and how HCI and HFE can navigate the territory together.

We are seeking position papers that provide commentary on AI challenges, questions, and opportunities from the lens of HCI and/or HFE. Submissions are welcome to focus on specific industries (e.g., aviation, cybersecurity, healthcare) and should be 2–4 pages in length. Accepted papers must have a dual focus on AI and HCI and/or HFE and will be assessed based on coherence, quality, and relevance to the workshop. Submissions should be sent to Alyssa Iglar (alyssa.iglar@mail.utoronto.ca) by February 19th, 2024 at 11:59 p.m. EDT. If accepted, we will request that authors extend their original submissions for possible inclusion in a proposed book that the organizers are seeking to publish through the Cambridge University Press, MIT Press, or Oxford University Press.

A hybrid workshop (website: <https://hcihfetogther.wordpress.com>) consisting of several activities and talks that afford in-person,

remote, and asynchronous engagement will be hosted. A minimum of one author for each accepted paper must be in attendance and register for the workshop as well as for one day or more of CHI 2024.

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